Lecture 8 - 17/06

Thursday, July 2, 2026 11:02 AM

limiting set -

for a trajetory $\varphi(t, x_0)$

$\{t_n\} : t_n \xrightarrow[n\to\infty]{} \infty$

$|\varphi(t_n, x_0) - x^*| \to 0$ then x^* is a limiting point

the set of limiting points is the limiting set $w(x_0)$

in a bounded trajectory, there must be a limiting set
in 2D, its either a FP or a cycle.

inside the cycle there must be a FP (or undefined area)

reason: going backwards in time, trajectories are spiraling inwards in a bounded area, they must approach a FP, or another smaller cycle (then repeat the argument)

example:
chemical reaction, 2 chemicals, X, Y, and more constants (basically infinite) chemicals A

converts

$\begin{cases} A \to X \\ 2X + Y \to 3X \\ B + X \to Y \\ X \to \emptyset \end{cases}$

$\dot{x} = a + x^2 y - bx - x$
$\dot{y} = bx - x^2 y$

$\dot{y} = 0 = bx - x^2 y$
$y = \frac{b}{x}$

$\dot{x} = 0 = a + x^2 y - bx - x$
$a + x^2 \frac{b}{x} - bx - x = 0$
$x = a$

FP at $(a, \frac{b}{a})$

$DF = \begin{pmatrix} 2xy - (1+b) & x^2 \\ b - 2xy & -x^2 \end{pmatrix}$

$DF(a, \frac{b}{a}) = \begin{pmatrix} b-1 & a^2 \\ & \end{pmatrix}$ $\det = \delta = -a^2(b-1) + ba^2 = -ba^2 + a^2 + ba^2 = a^2 > 0$

$$DF(a, \tfrac{b}{a}) = \begin{pmatrix} b-1 & a^2 \\ -b & -a^2 \end{pmatrix} \qquad \det = \Delta = -a^2(b-1) + ba^2 = -ba^2 + a^2 + ba^2 = a^2 > 0$$

$$tr = \tau = b - 1 - a^2 = b - (1+a^2)$$

if $b > 1+a^2$ → repellar

if $b < 1+a^2$ → attractor

lets look at $b > 1+a$

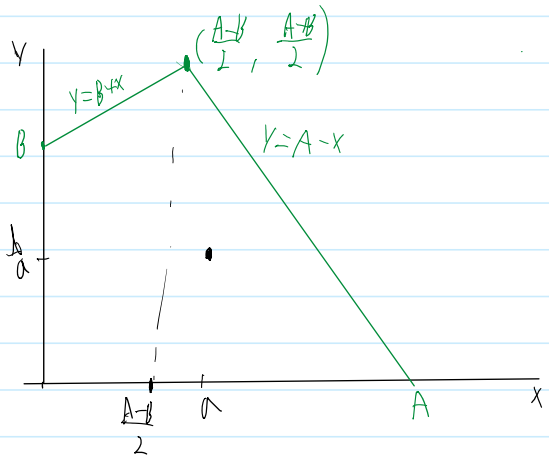
it we show its bounded, then there will be a limit cycle, the repelled trajectories will go to.

lets show its bounded

for $Y=0$, $\dot{Y} = bX > 0$,

for $X=0$ $\dot{X} = a > 0$

so you stay in the first quadrant



lets guess the following boundry - $Y = B+X$, $Y = A-X$
where $B < A$

for $Y = A - X$

define $L(x,y) = x+y-A$, basically the distance from the line (below it is negative, above, positive)

$\dot{L} = \dot{x}+\dot{y} = a + \cancel{x^2y} - bx - x + bx \cancel{- x^2y} = a - x$

for $\frac{A-B}{2} < x < A$ for $\dot{L} < 0$, we need $a < x$, as $\frac{A-B}{2}$ is the smallest relevant x, $\boxed{\frac{A-B}{2} > a}$

for $Y = B+X$

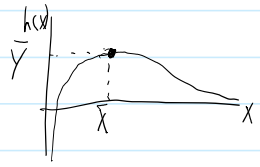
$L(x,y) = y - x - B$

$\dot{L} = \dot{y} - \dot{x} = bx - x^2y - a - x^2y + bx + x = 2bx - 2x^2y + x - a = (2b+1)x - 2x^2y - a$

$\dot{L} < 0$, $2bx - 2x^2y + x - a < 0$

$2bx + x - a < 2x^2y$

$h(x) = \frac{(2b+1)x - a}{2x^2} = \frac{2bx + x - a}{2x^2} < y$



$h(x)$ has some max, lets find it

$h'(x) = \frac{-(2b+1)x + 2a}{2x^3}$

$h'(x) = 0$, $0 = -(2b+1)\bar{X} + 2a$

$\bar{X} = \frac{2a}{2b+1}$

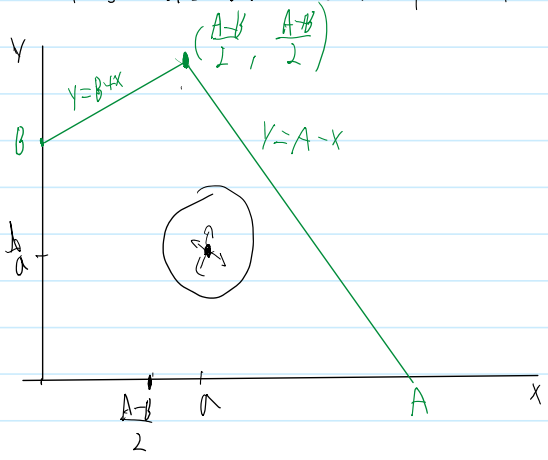
$\bar{Y} = h(\bar{X}) = \frac{\cancel{(2b+1)} \cdot \frac{2a}{\cancel{2b+1}} - a}{2 \cdot \frac{4a^2}{(2b+1)^2}} = \frac{a(2b+1)^2}{8a^2} = \frac{(2b+1)^2}{8a}$

$Y = h(\bar{X}) = \frac{\tan\gamma \cdot 2\pi r - 0}{2 \cdot \frac{4\alpha^2}{(2n+1)^2}} = \frac{\alpha x \cdot (2n+1)}{8\alpha^2} = \frac{(2n+1)^2}{8\alpha}$

if $Y > \bar{Y} = \frac{(2n+1)^2}{8\alpha}$ then $L' < 0$. $\quad Y > B > \bar{Y}$

the smallest Y there, is B, so if $Y \geq B > \bar{Y}$, L' is negative

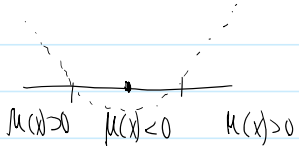
So the trajectories are bounded, so ther must be an orbit within it



example: we had harmonic oscilator with friction

$\ddot{x} = -x - \mu\dot{x}$

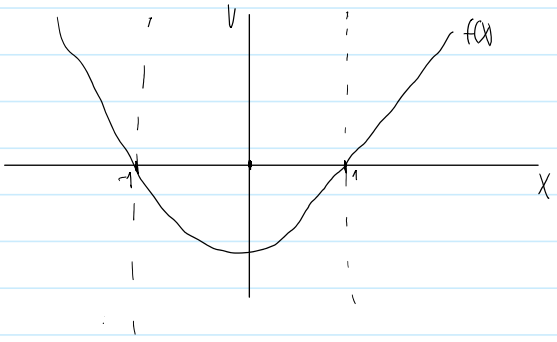
lets look at $\mu(x)$ depends on x and can be < 0



$\mu(x) = x^2 - 1$

$\dot{x} = v$

$\dot{v} = -x - \overbrace{(x^2-1)}^{f(x)} \cdot v$



define

$Y = V + \int_0^x f(t)dt = V + \overbrace{\frac{x^3}{3} - x}^{F(x)}$

rewrite the system to x, Y

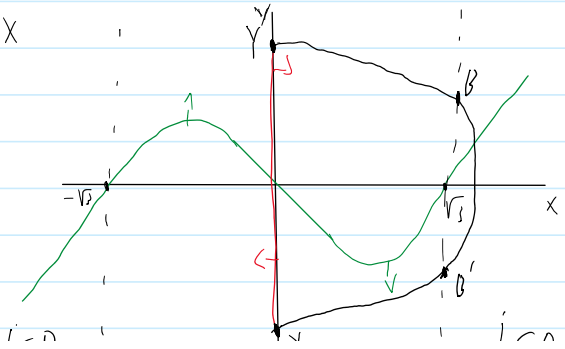
$\dot{x} = Y - F(x)$

$\dot{y} = \dot{v} + F'(x) \cdot \dot{x} = -x - \cancel{f(x) \cdot v} + \cancel{f(x)} v = -x$

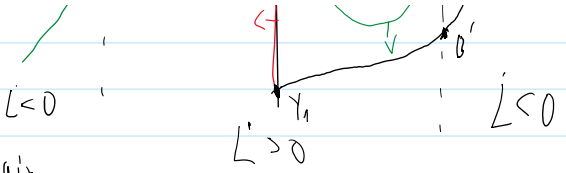
nullclines

$\dot{x} = 0$, $Y = F(x)$

$\dot{Y} = 0$, $X = 0$



$\dot{Y}=0$, $X=0$



$L(X,Y)=\frac{X^2}{2}+\frac{Y^2}{2}$ distance from origin

$L'=X\dot{X}+Y\dot{Y}=XY-XF(X)-YX=-XF(X)$

So if we start at Y we'll get to Y_1 we want to see if $L(Y_1)\overset{?}{\leq}L(Y)$ then we'll know the shape of the cycle (or spiral)

$\Delta L(Y)=L(Y_1)-L(Y)$

for small Y, $\Delta L(Y)>0$, if the whole way from Y to Y_1 is within $(-\sqrt{3},\sqrt{3})$

for other Y_1 we have 3 parts, start from $L'>0$, cross to $L'<0$, then again $L'>0$

lets say at t_Y we hit $X(t_Y)=\sqrt{3}$, lets call that point B

so $\Delta L(Y)=L(Y_1)-L(Y)=\underbrace{L(B)-L(Y)}_{>0}+\underbrace{L(B')-L(B)}_{<0}+\underbrace{L(Y_1)-L(B')}_{>0}$

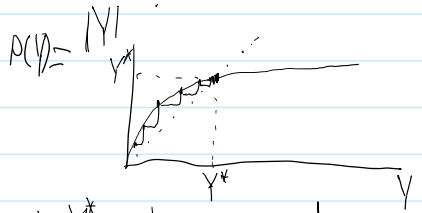
$L(B)-L(Y)=\int_0^{t_Y}\dot{L}\,dt=\int_0^{t_Y}-XF(X)\,dt=\int_0^{t_Y}\dot{Y}F(X)\,dt=\int_Y^{Y_B}dY\cdot F(X)=\int_0^{Y_B}\frac{-X}{Y-F(X)}F(X)\xrightarrow{Y\to\infty}0$

$\dot{Y}dt=dY$

$\frac{dY}{dX}dX=\frac{\dot{Y}}{\dot{X}}\cdot dX=\frac{-X}{Y-F(X)}$

$L(Y_1)-L(B)=$ similar to above, $\xrightarrow[Y\to\infty]{}0$

$|L(B')-L(B)|$ increases with greater Y (bad examination)



so there is some point Y^* where we end up where we started, that's a cycle. anything below it, spiral outwards to it and vice versa

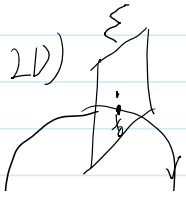
$P(Y^*)=Y^*$

if $P'(Y^*)<1$ → stable periodic orbit

in the general case we have a cycle flow (can be more than 2D)

$X_0\xrightarrow{T}\varphi(T;X_0)=X_0$

take a plane that is perpendicular to the flow Σ

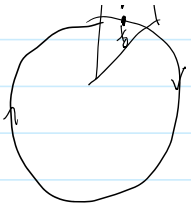


$x_0 \xrightarrow{T} \varphi(T; x_0) = x_0$

take a plane that is perpendicular to the flow Σ

take a point close to the cycle, because its close and continuous

it'll travel with the cycle, so after T-period, it'll cross the plane



$x \in \Sigma \xrightarrow{T' \approx T} P(T'; x) = x' \in \Sigma \quad = P(x)$ called a Poincare map

this function is of dim $D-1$

in 2D we can draw it, and the $P'(x^*)$ determins if its stable or not

$P(x_0) = x_0$

$x = x_0 + \delta x \to x' = P(x_0 + \delta x) = P(x_0) + P(\delta x) \overset{\text{linearize}}{=} x_0 + DP(x_0) \cdot \delta x_0$

$\parallel$ ↑ partial drivative matrix

in 2D, this is $P'(x)\,\delta x_0$; if $P'(x) < 1$, it'll shrinks

in otder, all eigen values need to be < 1

Bifurcation

our systems have parameters, usually a slight change in them leads to a small change in the systems. sometimes crossing some value can lead to dramatic changes

before we saw 2 chemicals a, b, and we said $A > 1+a$

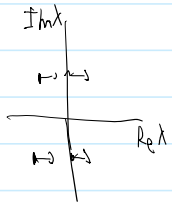
so crossing this creates the cycle, basically the system has the cycle potential, but without the right params, its not realized, it has an imaginary part eigenvalue

this is called Hopf bifurcation

example:

$$\dot{x} = \mu x + kx(x^2+y^2)$$
$$\dot{y} = wx + \mu y + ky(x^2+y^2)$$

linearizing, we get $A_{(0,0)} = \begin{pmatrix} \mu & -w \\ w & \mu \end{pmatrix}$



$(\mu-\lambda)^2 + w^2 = \lambda^2 - 2\mu\lambda + w^2$

$\lambda_{1,2} = \mu \pm \sqrt{\mu^2 - w^2}$

look at $\mu \approx 0$

$\lambda_{1,2} \approx \mu \pm iw$

raising μ makes it cross from negative to positive

so critical point is $\mu_{cr} = 0$

to critical point is $\mu_{cr} = 0$

lets switch to polar coords

$x = r\cos\theta$
$y = r\sin\theta$

$\dot{x} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta$
$\dot{y} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta$

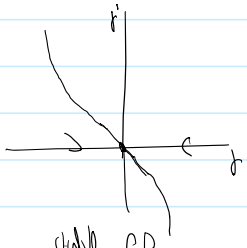
$\Downarrow$ in our case (after some substitution)

$\dot{r} = \mu r + Kr^3$
$\dot{\theta} = w$

because $\dot{\theta} = w$, we are rotating clockwise
to know spiral direction, look at $\dot{r}$

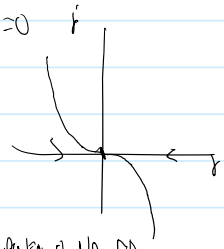
① $K < 0$ supercritical Hopf

ⓐ $\mu < 0$



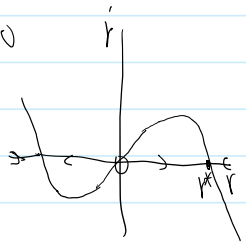
stable FP

ⓑ $\mu = 0$



weakly stable FP

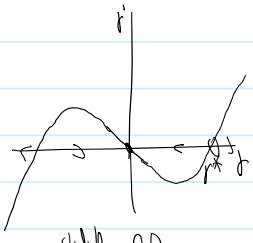
ⓒ $\mu > 0$



unstable FP
stable PO

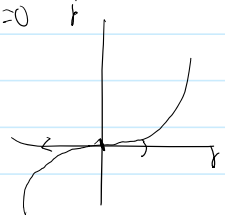
② $K < 0$ subcritical Hopf

ⓐ $\mu < 0$



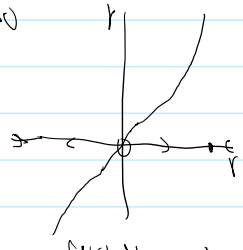
stable FP
r^* unstable PO

ⓑ $\mu = 0$



weakly unstable

ⓒ $\mu > 0$



unstable FP

this is a general thing

if $\lambda_{1,2} = \alpha_\mu \pm i w_\mu$

① $\alpha_{\mu_{cr}} = 0$ the real part crosses from negative to positive

$\alpha'_{\mu_{cr}} > 0$

look at it when $\mu = \mu^*$ ($\mu = 0$ in our case)
weakly stable $\Rightarrow$ supercritical $\Rightarrow$ PO appear after bifurcation
weakly unstable $\Rightarrow$ subcritical $\Rightarrow$ PO appear before "

weakly stable ⇒ supercritical ⇒ PO appear after bifurcation
weakly unstable ⇒ subcritical ⇒ PO appear before "

example

$$\begin{cases} \dot{x} = y \\ \dot{y} = -x - x^3 + \mu y + k x^2 y \end{cases}$$

FP at $(0,0)$

taking partial derivative of linear part

$$A = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}$$

$$-\lambda(\mu - \lambda) + 1 = 0$$

$$\lambda^2 - \mu\lambda + 1 = 0$$

$$\lambda_{1,2} = \frac{\mu \pm \sqrt{\mu^2 - 4}}{2} \overset{\text{when } \mu = 0}{=} \frac{\mu}{2} \pm i \cdot 1$$

we have a Hopf bifurcation, but which one?

look at the system at bifurcation point $\mu = 0$

$$\dot{x} = y$$
$$\dot{y} = -x - x^3 + k x^2 y$$

$$L(x,y) = \frac{x^2}{2} + \frac{x^4}{4} + \frac{y^2}{2}$$

$$\dot{L} = x\dot{x} + x^3 \cdot \dot{x} + y\dot{y} = \cancel{(x + x^3)y} + y(\cancel{-x - x^3} + k x^2 y) = k x^2 y^2 \geq 0$$

so FP is weakly unstable ⇒ subcritical Hopf

so there is an unstable PO at $\mu < 0$